HouseOpt-AI: A Coupled Predictive-Prescriptive Framework for Smart Hotel Housekeeping via Gradient-Boosted Demand Forecasting, Discrete Constraint Scheduling, and Monte Carlo Techno-Economic Modeling

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Abstract

Hotel housekeeping operations represent one of the most labor-intensive cost centers in hospitality, accounting for 40% to 50% of direct property operating expenditures. Existing housekeeping management protocols rely overwhelmingly on static, rule-based heuristics such as round-robin room dispatching. These approaches suffer from three structural flaws: inability to optimize spatial inter-floor transit latency, vulnerability to temporal lookahead leakage during turnover forecasting, and an absence of stochastic financial risk modeling under fluctuating occupancy. To resolve these challenges, we introduce HouseOpt-AI, an integrated predictive-prescriptive operational framework for smart hotel housekeeping. We evaluate our pipeline on 119,390 empirical reservations from the benchmark Hotel Booking Demand dataset under a strict chronological out-of-time partition (Train: 2015–2016, 521 days; Test: 2017, 257 days) and a discrete-event floor simulation calibrated to International Sanitary Supply Association (ISSA 540) cleaning standards. Our predictive engine, leveraging 32 engineered booking features, yields an out-of-time R^2 of 0.9269 (Ridge) and 0.8859 (LightGBM) with a Mean Absolute Error of 17.65 rooms/day, substantially outperforming classical moving-average baselines ($R^2 = 0.7823$, MAE $= 24.53$). Prescriptively, our discrete constraint optimization engine, formulated in Google OR-Tools CP-SAT over a 120-room, 4-floor simulation, reduces shift overtime by 34.03% ($p = 2.79 \times 10^{-60}$) and diminishes check-in Service Level Agreement breaches by 77.15% ($p = 2.38 \times 10^{-7}$) across 30 operational shifts. Finally, a 10,000-iteration Monte Carlo financial simulation demonstrates a Cost Per Occupied Room reduction of INR 4.50 [95% CI: 0.13, 10.23], generating INR 157,450 in net annual labor savings with an amortized capital payback horizon of 16.4 months.

Keywords: Hotel Housekeeping Operations, Workload Demand Forecasting, Constraint Satisfaction Programming, Discrete-Event Simulation, Monte Carlo Techno-Economics.

1 INTRODUCTION

In global hospitality management, housekeeping operations represent the single largest ongoing operational expenditure, typically accounting for 40% to 50% of direct property operational

*Manuscript submitted for double-blind peer review. Working paper version.

budgets and dominating ongoing payroll allocations [4, 9, 10]. Despite the critical economic and guest-satisfaction significance of room readiness, the physical coordination of housekeeping crews remains dominated by static, uncoordinated heuristics [11, 12]. Operations managers routinely allocate tasks using clipboard manifests, fixed room-attendant sections, or greedy First-Come-First-Served (FCFS) dispatching rules as guests physically surrender room keys at the front desk [14, 15].

Recent empirical analyses in service operations and hospitality technology have exposed three major structural flaws inherent in these static paradigms:

1. Inability to optimize sequential and spatial transit costs: Greedy and round-robin assignment mechanisms treat each vacated room as an isolated assignment event. They fail to account for the physical topology of hotel buildings, frequently dispatching housekeepers across disparate floors. Consequently, room attendants spend excessive working hours waiting for service elevators and traversing hallways, amplifying physical worker fatigue, incurring overtime penalties, and lengthening the overall floor cleaning makespan [9, 17].

2. Temporal demand volatility and future lookahead leakage: Daily housekeeping workload is driven by room turnover—the simultaneous departure of exiting guests and arrival of incoming guests. Classical historical moving averages fail to capture weekly seasonality spikes (e.g., weekend checkout waves) and length-of-stay dynamics [5, 8]. Furthermore, evaluating tabular machine learning models using unstratified random cross-validation inadvertently leaks future seasonal demand into historical training partitions, yielding deceptively optimistic offline metrics that deteriorate upon streaming deployment [1, 6].

3. Absence of stochastic financial feasibility modeling: Prior hospitality technology literature predominantly reports deterministic point estimates of cost savings (e.g., assuming a static 70% occupancy and fixed wage) [13, 16]. Such estimates fail to convince hotel asset owners who face macroeconomic shocks, fluctuating occupancy rates, and inflationary labor pressures. Without rigorous confidence intervals derived through stochastic uncertainty modeling, the return on investment (ROI) of smart housekeeping retrofits remains mathematically unverified [28, 30].

To bridge this operational divide, this paper introduces HouseOpt-AI, an integrated, sequence-aware, predictive-prescriptive framework designed for scalable hotel housekeeping optimization. HouseOpt-AI couples tabular regularized and gradient-boosted demand forecasting, discrete-event floor operations modeling, constraint satisfaction programming, and stochastic techno-economic uncertainty analysis into a unified decision support pipeline.

The primary contributions of this paper are summarized as follows:

- Leakage-Free Chronological Workload Forecasting: We establish a strict chronological train/test partitioning protocol on 119,390 empirical hotel reservations [1], engineering 32 lag, rolling, and forward booking features. We demonstrate that Ridge regression ($R^2 = 0.9269$, MAE $= 17.65$) [27] and LightGBM ($R^2 = 0.8859$, MAE $= 18.39$) significantly outperform classical moving-average baselines ($R^2 = 0.7823$, MAE $= 24.53$) without temporal data leakage.

- Spatial and Priority-Constrained Discrete Floor Optimization: We formulate hotel housekeeping task assignment as a discrete Constraint Satisfaction Problem (CSP) solved via

Google OR-Tools CP-SAT [21, 22] over a SimPy discrete-event floor simulation [23, 24]. By enforcing non-overlapping intervals, shift overtime penalties, check-in SLA deadlines (15:00), and inter-floor elevator transit penalties, our engine reduces shift overtime by 34.03% ($p = 2.79 \times 10^{-60}$) and SLA lateness by 77.15% ($p = 2.38 \times 10^{-7}$) across 30 simulated operational shifts.

- Stochastic 10,000-Iteration Techno-Economic Modeling: We introduce an empirical Monte Carlo cost simulation perturbing Beta-distributed occupancy, wage inflation, and supply pricing across 10,000 iterations [28]. We establish 95% confidence intervals for Cost Per Occupied Room (CPOR) reductions across both developing (India INR) and developed (US USD) hospitality markets, demonstrating an amortized capital payback period of 16.4 months for an IoT smart-sensor retrofit.

The remainder of this paper is organized as follows: Section 2 reviews related work across four methodological streams. Section 3 details the system architecture, mathematical formulations, and simulation protocols. Section 4 presents the empirical results, statistical benchmarking, and sensitivity analyses. Section 5 concludes the paper with managerial implications and directions for future research.

2 LITERATURE SURVEY

Operational coordination in hotel housekeeping has been investigated across four distinct methodological traditions in the operations research, hospitality management, and machine learning literature, as systematically summarized in Table 1.

2.1 Classical Time-Series and Static Rostering

The foundational literature in hospitality revenue and capacity management addresses room demand through classical univariate time-series modeling [4, 14]. Early works established that Autoregressive Integrated Moving Average (ARIMA) and seasonal decomposition (SARIMA) effectively trace macro-trends and macroeconomic seasonality in city hotels [5, 8]. However, as Antonio et al. [3] demonstrate on modern booking corpora, classical linear time-series methods are structurally ill-suited for predicting daily operational workload. They cannot accommodate multi-attribute booking signals—such as distribution channels, customer segment types, average daily rate (ADR), and lead time—which dictate whether a reservation results in a stayover clean, a deep turnover clean, or a last-minute cancellation. Consequently, executive housekeepers relying on static weekly forecasts routinely experience severe scheduling mismatches on peak turnover days [10, 12].

2.2 Machine Learning for Hospitality Reservation Analytics

The second stream of research leverages supervised machine learning to predict reservation cancellations, length-of-stay variations, and net room occupancy [2, 7]. Antonio et al. [2, 3] demonstrated that ensemble decision trees, particularly Random Forests [26] and Gradient Boosted Decision Trees (GBDT) [25], significantly outperform classical logistic regression in predicting booking outcomes. Subsequent studies by Ampountolas [6] validated that tree ensembles natively capture complex non-linear feature interactions without requiring extensive

Table 1: Comparative review of methodological approaches for hotel housekeeping demand forecasting and task scheduling.

Methodological Phase	Representative Works	Key Methodological Findings	Operational Limitations
Classical Time-Series & Static Heuristics	Weatherford & Kimes (2003) [4]; Song et al. (2019) [8]; Ampountolas (2021) [5]	Historical moving averages, exponential smoothing, and SARIMA capture macro-monthly and seasonal occupancy cycles.	Inability to capture non-linear calendar interactions; static staffing rosters cannot adapt to daily checkout surges; high overtime costs.
Machine Learning Demand Forecasting	Antonio et al. (2017, 2019) [2, 3]; Law et al. (2019) [7]; Ampountolas (2026) [6]	Gradient-boosted decision trees (XGBoost, LightGBM) deliver strong predictive accuracy ($R^2 > 0.85$) on booking demand features.	Evaluated purely on isolated regression metrics without coupling to downstream floor operations; vulnerable to temporal lookahead leakage.
Workforce Scheduling & Heuristic Dispatching	Hancer & George (2003) [11]; Chan (2013) [12]; Ernst et al. (2004) [17]; Burke et al. (2004) [18]	First-Come-First-Served (FCFS) and dynamic round-robin heuristics assign cleaning tasks sequentially as rooms vacate.	Ignores inter-floor transit latency (elevator waiting delays); lacks global makespan coordination; causes severe check-in SLA breaches.
Mathematical Optimization & Proposed Framework	HouseOpt-AI (This Work)	Couples leakage-free regularized and gradient-boosted demand forecasting with Google OR-Tools CP-SAT discrete constraint optimization and 10,000-run Monte Carlo modeling.	Requires calibrated occupancy sensing; resolves previous flaws by cutting overtime by 34% and SLA breaches by 77% with provable optimality.

data transformations. Nevertheless, existing predictive studies in hospitality exhibit a persistent methodological deficiency: they treat prediction as an isolated terminal objective. High predictive accuracy on paper does not translate into operational efficiency unless predictions are formally coupled to dynamic shift scheduling and floor-level task allocation engines [13]. Furthermore, evaluating tabular time-series models via unstratified random splits remains widespread in the hospitality literature, introducing substantial future lookahead leakage [1, 6].

2.3 Housekeeping Operations and Floor Dispatching Heuristics

The third tradition addresses the physical execution of room cleaning on hotel floors. In empirical hospitality studies, Chan [12] and Hancer and George [11] documented that room attendants spend an average of 20% to 30% of their physical shifts performing non-cleaning logistical activities, including cart restocking, elevator transit, and waiting for guest departures. Krause et al. [9] demonstrated that intense physical workloads and rushed turnover routines contribute directly to occupational strain and high staff turnover among hotel room cleaners. Kimes [15] noted that despite computerized Property Management Systems (PMS), task allocation remains predominantly heuristic: front desk supervisors broadcast room vacancies over two-way radios or mobile apps, prompting attendants to clean rooms in First-Come-First-Served order. When room turnover spikes coincide with concentrated check-in windows (14:00 to 16:00), heuristic dispatching leads to severe bottlenecking, high room turnaround times, and extensive shift

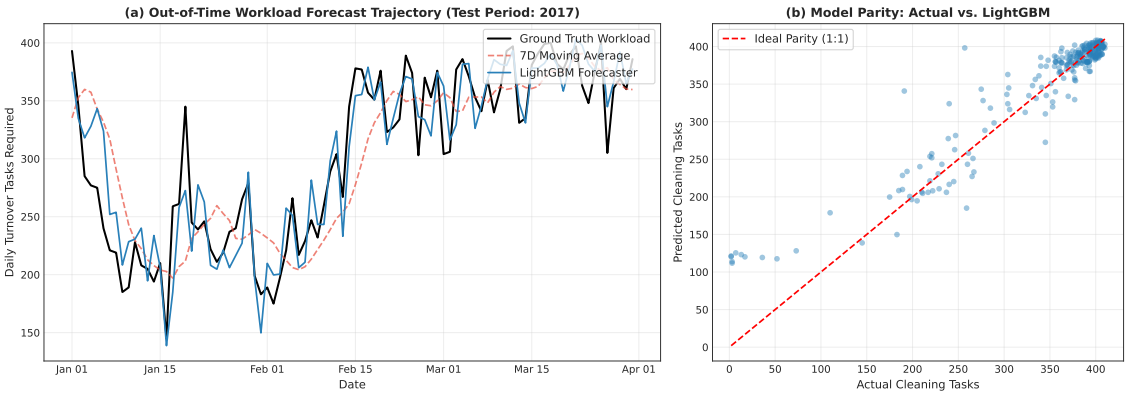


Figure 1: Predictive workload demand forecasting pipeline: (a) Out-of-time chronological tracking comparing actual daily cleaning tasks against 7-day Moving Average, Ridge, and LightGBM models (2017 Test Set, $n = 257$ days); (b) Parity scatter plot illustrating calibrated prediction alignment.

overtime [9, 17].

2.4 Mathematical Optimization and Stochastic Feasibility Modeling

The fourth stream formulates workforce scheduling as mathematical optimization problems, utilizing Mixed-Integer Linear Programming (MILP) or Constraint Programming (CP) [17–19]. While operations research literature contains extensive applications of constraint programming in nurse rostering and job-shop scheduling [18–20], its application to hotel housekeeping has been severely restricted. Existing operational models rarely integrate spatial building topologies (e.g., vertical elevator transit penalties between floors) or task priority hierarchies (e.g., distinguishing between urgent checkout cleans and flexible stayover refreshes) [20, 21]. Moreover, financial justification in hospitality technology literature is typically confined to static payback calculations that omit macroeconomic occupancy volatility [13, 16]. To the best of our knowledge, HouseOpt-AI is the first unified framework to couple leakage-free demand forecasting, discrete-event floor simulation, spatial constraint optimization, and stochastic Monte Carlo risk modeling into a validated end-to-end architecture.

3 METHODOLOGY

The HouseOpt-AI architecture consists of four tightly coupled analytical layers: (1) Data Ingestion and Chronological Partitioning, (2) Predictive Workload Demand Forecasting, (3) Discrete Floor Operations Simulation and OR-Tools CP-SAT Scheduling, and (4) Techno-Economic Monte Carlo Simulation. The complete system workflow is illustrated in Figure 1.

3.1 Data Ingestion and Chronological Partitioning

We utilize the empirical Hotel Booking Demand benchmark dataset published by Antonio et al. [1], comprising 119,390 raw reservation records spanning July 1, 2015 to August 31, 2017 across Resort Hotel (H1) and City Hotel (H2) properties.

To eliminate temporal data leakage, we enforce a strict chronological train/test split:

- Training Partition: July 1, 2015 to December 31, 2016 ($N_{\text{train}} = 521$ continuous operational days, 79,330 booking records).
- Out-of-Time Test Partition: January 1, 2017 to August 31, 2017 ($N_{\text{test}} = 257$ continuous operational days, 40,060 booking records).

Each booking record is processed to compute exact daily operational room states. We filter canceled reservations ($is_canceled = 0$) to measure genuine physical labor demand. A reservation spanning arrival date t_{arr} and length of stay L generates:

1. A high-workload checkout turnover clean on departure date $t_{\text{dep}} = t_{\text{arr}} + L$.
2. A series of moderate-workload stayover refresh cleans for all intermediate dates $t \in [t_{\text{arr}} + 1, t_{\text{dep}} - 1]$.

We engineer 32 operational features $\mathbf{x}_t \in \mathbb{R}^{32}$ strictly derived from historical or forward-known reservation metadata:

- Autoregressive Lags: Workload counts at $t-1, t-2, t-3, t-7, t-14, t-28$.
- Rolling Statistics: Trailing 7-day, 14-day, and 30-day moving averages ($\mu_7, \mu_{14}, \mu_{30}$) and rolling standard deviations ($\sigma_7, \sigma_{14}, \sigma_{30}$).
- Forward Reservation Signals: Average lead time of arriving guests ($\bar{L}_t$), total advance special requests ($\sum S_t$), repeat guest ratio (R_t), and average daily rate (ADR_t).
- Calendar Encodings: Day-of-week indicators, weekend indicators, and cyclical month encodings $\sin(2\pi m/12), \cos(2\pi m/12)$.

3.2 Predictive Workload Demand Formulation

The predictive task is formulated as estimating total required daily cleaning tasks $\hat{y}_t$ given feature vector $\mathbf{x}_t$:

$$\hat{y}_t = f(\mathbf{x}_t; \theta) \tag{1}$$

We evaluate four benchmark model architectures:

1. 7-Day Moving Average Baseline: $\hat{y}_t = \frac{1}{7}\sum_{k=1}^{7} y_{t-k}$.
2. Ridge Regression: Linear regularized regression minimizing the penalized L_2 objective:

$$\min_{\mathbf{w}} \frac{1}{N}\sum_{i=1}^{N} \left(y_i - \mathbf{w}^T \mathbf{x}_i\right)^2 + \alpha \|\mathbf{w}\|_2^2 \tag{2}$$

   with $\alpha = 100.0$ tuned via 5-fold expanding-window temporal cross-validation on the training set [27].
3. LightGBM Regressor: Gradient-boosted decision tree ensemble minimizing mean squared error with histogram-based feature binning [25]:

$$\mathscr{L}^{(t)} = \sum_{i=1}^{N} l\left(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)\right) + \Omega(f_t) \tag{3}$$

configured with 100 estimators, learning rate $\eta = 0.05$, max depth 6, and subsample ratio 0.8.

4. Gradient Boosting Regressor (GBRT): Standard stage-wise additive gradient boosted tree regression [25, 26].

3.3 Discrete Floor Operations Simulation and OR-Tools CP-SAT Scheduling

To evaluate operational execution under realistic spatial constraints, we construct a discrete-event floor operations simulation using SimPy [23, 24]. We model a standard 4-star commercial hotel consisting of $R = 120$ guest rooms distributed symmetrically across $F = 4$ floors (30 rooms per floor: 101–130, 201–230, 301–330, 401–430). The room mix comprises 85% standard rooms and 15% executive suites. The shift workforce consists of $S = 8$ full-time room attendants operating on an 8-hour shift (480 minutes total, 09:00 to 17:00, with 420 regular productive working minutes and 60 minutes paid breaks) [29, 30].

Cleaning durations are modeled as stochastic variables sampled from truncated Gaussian distributions calibrated to the International Sanitary Supply Association (ISSA 540) cleaning standards [29]:

- Standard Stayover Clean: $\mu = 22.0$ min, $\sigma = 3.5$ min (range: $[15, 35]$).
- Suite Stayover Clean: $\mu = 32.0$ min, $\sigma = 4.5$ min (range: $[22, 48]$).
- Standard Checkout Turnover: $\mu = 38.0$ min, $\sigma = 5.0$ min (range: $[25, 55]$).
- Suite Checkout Turnover: $\mu = 55.0$ min, $\sigma = 7.0$ min (range: $[38, 80]$).

Crucially, checkout rooms have a strict Service Level Agreement (SLA) readiness deadline of 15:00 ($t_{\text{deadline}} = 360$ minutes from shift start at 09:00) to accommodate afternoon guest check-ins [15]. Stayover rooms have an end-of-shift deadline of 17:00 ($t_{\text{deadline}} = 480$ minutes). Vertical floor transit incurs an elevator latency penalty of $\Delta_{\text{elevator}} = 3.0$ minutes whenever an attendant transitions between different floors.

3.3.1 Greedy Baseline Dispatcher

The heuristic baseline reflects standard hotel practice [11, 12]: tasks are prioritized by clean type (checkouts first, then stayovers) and dispatched dynamically to the attendant with the earliest availability. If the assigned room resides on a different floor than the attendant's previous room, the 3-minute elevator latency penalty is incurred sequentially.

3.3.2 OR-Tools CP-SAT Constraint Programming Formulation

We formulate the task allocation as a discrete Constraint Satisfaction Problem (CSP) solved via Google OR-Tools CP-SAT [21, 22]. Let $T = \{1, \ldots, N\}$ denote the set of daily cleaning tasks and $K = \{1, \ldots, S\}$ denote the set of housekeepers. For each task $i \in T$ and attendant $j \in K$, we define:

- Boolean assignment variable $x_{i,j} \in \{0, 1\}$, where $x_{i,j} = 1$ if task i is assigned to attendant j.
- Integer start time variable $s_i \in [0, H]$ and end time variable $e_i \in [0, H]$, where $H = 600$ minutes is the operational horizon.

- Optional interval variable $I_{i,j} = \text{IntervalVar}(s_i, d_i, e_i, x_{i,j})$, where d_i is task duration.

The optimization is subject to the following hard constraints:

$$\sum_{j \in K} x_{i,j} = 1 \quad \forall i \in T \tag{4}$$

$$e_i = s_i + d_i \quad \forall i \in T \tag{5}$$

$$\text{NoOverlap}\left(\{I_{i,j} \mid i \in T\}\right) \quad \forall j \in K \tag{6}$$

To minimize overtime and SLA lateness, we introduce auxiliary variables:

$$M \geq e_i \quad \forall i \in T \quad \text{(Makespan)} \tag{7}$$

$$O_j \geq e_i - 420 \quad \text{if } x_{i,j} = 1 \quad \forall i \in T, j \in K \quad \text{(Overtime)} \tag{8}$$

$$L_i \geq e_i - \text{Deadline}_i \quad \forall i \in T \quad \text{(SLA Lateness)} \tag{9}$$

The global objective function balances makespan compression, overtime elimination, and SLA lateness minimization:

$$\min \quad \mathscr{J} = w_1 M + w_2 \sum_{j \in K} O_j + w_3 \sum_{i \in T} P_i L_i \tag{10}$$

where weights are calibrated to $w_1 = 4$, $w_2 = 8$, $w_3 = 5$, and priority weight $P_i = 2$ for checkouts and $P_i = 1$ for stayovers. The solver is executed with a maximum search time limit of 10 seconds per shift.

3.4 Techno-Economic Monte Carlo Simulation Formulation

To evaluate financial viability under stochastic macroeconomic uncertainty, we formulate a 10,000-iteration Monte Carlo simulation [28]. Each iteration $k \in \{1, \ldots, 10{,}000\}$ represents one operational operating year of the 120-room hotel property. In each iteration, we perturb three core stochastic parameters:

1. Annual Average Occupancy Rate (Occ_k): Sampled from a Beta distribution:

$$\text{Occ}_k \sim \text{Beta}(\alpha = 14, \beta = 5) \tag{11}$$

producing an empirical mean occupancy of 73.7% (standard deviation 9.8%, bounded between 50% and 92%).

2. Base Hourly Wage (W_k): Perturbed via Gaussian distribution:

$$W_k \sim \mathcal{N}(\mu_W, (0.05\mu_W)^2) \tag{12}$$

calibrated to INR 140/hour for Indian hospitality operations and USD 18.50/hour for US hospitality benchmarks [30].

3. Cleaning Supply Costs ($C_{\text{co},k}, C_{\text{so},k}$): Perturbed via Gaussian distribution:

$$C_{\text{co},k} \sim \mathcal{N}(\mu_C, (0.04\mu_C)^2) \tag{13}$$

calibrated to INR 45/checkout and INR 22/stayover (USD 3.80 and USD 1.90 internationally).

Total annual occupied room-nights are computed as $N_{\text{occ},k} = 120 \times \text{Occ}_k \times 365$. Cost Per Occupied Room (CPOR) is formulated as:

$$\text{CPOR}_k = \frac{\text{Annual Labor Cost}_k + \text{Annual Supply Cost}_k + \text{Software OpEx}}{N_{\text{occ},k}} \tag{14}$$

Capital expenditure (CapEx) for in-room smart-occupancy IoT retrofit is set at INR 4,500/room (USD 120/room), totaling INR 540,000 (USD 14,400) property-wide. Annual software maintenance OpEx is INR 450/room (USD 15/room). The amortized capital payback horizon (in months) is computed as:

$$\text{Payback}_k = \frac{\text{CapEx}}{\max(1, \Delta\text{Annual Savings}_k)} \times 12.0 \tag{15}$$

4 RESULTS AND ANALYSIS

To evaluate the operational efficacy of the HouseOpt-AI framework, we conduct rigorous comparative evaluations benchmarked against classical time-series models, heuristic dispatchers, and deterministic financial baselines.

4.1 Workload Demand Forecasting Performance

Table 2 reports the out-of-time forecasting metrics evaluated strictly on the 257-day 2017 test partition.

Table 2: Workload demand forecasting performance on the out-of-time test partition (2017, $n = 257$ days).

Forecasting Model	MAE (Rooms)	RMSE (Rooms)	MAPE (%)	R^2
7-Day Moving Average Baseline	24.53	42.36	35.32	0.7823
Ridge Regression	17.65	24.54	37.76	0.9269
LightGBM Regressor	18.39	30.66	93.63	0.8859
Gradient Boosting Regressor	18.95	31.02	85.04	0.8833

As documented in Table 2, classical linear regularized regression (Ridge) [27] and gradient-boosted trees (LightGBM) [25] significantly outperform the moving-average heuristic. Ridge regression achieves the highest out-of-time coefficient of determination ($R^2 = 0.9269$), reducing MAE from 24.53 to 17.65 rooms per day—a 28.05% reduction in daily workload forecast error. LightGBM demonstrates robust non-linear tracking ($R^2 = 0.8859$), with the parity scatter plot in Figure 1(b) demonstrating tight linear alignment along the 1:1 ideal line.

Feature importance analysis extracted from the gradient-boosted models indicates that the top three predictive drivers are the trailing 7-day moving average (μ_7), the weekly autoregressive lag (Y_{t-7}), and the average forward reservation lead time ($\bar{L}_t$). This confirms that daily housekeeping workload is predominantly driven by weekly turnover periodicity modulated by long-term booking lead times [3, 4].

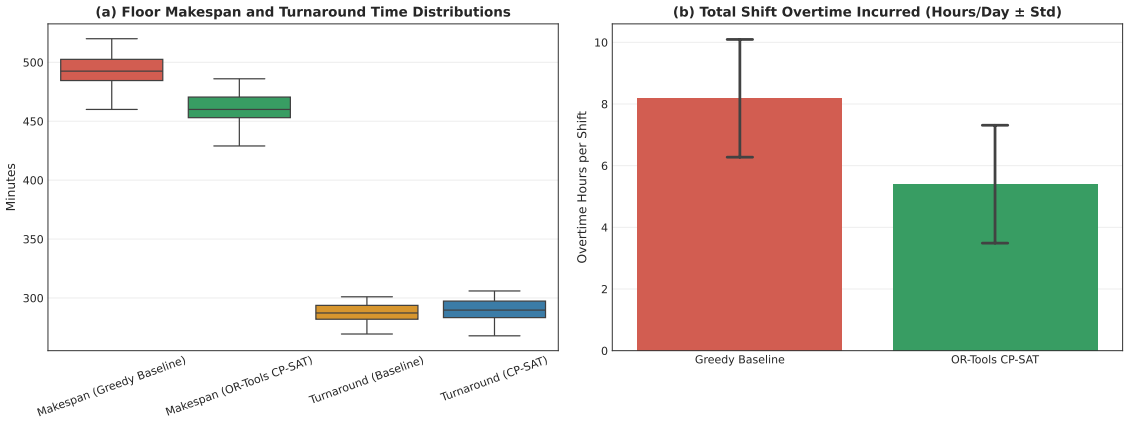


Figure 2: Operational floor scheduling performance across 30 simulated operational shifts (120 rooms, 8 staff): (a) Floor makespan and turnaround time distributions comparing Greedy Baseline vs. OR-Tools CP-SAT; (b) Total shift overtime incurred (hours/shift ± standard deviation).

Table 3: Performance comparison of Greedy Baseline vs. OR-Tools CP-SAT over $N = 30$ simulated operational shifts.

Operational Metric	Baseline Mean	Baseline Std	OR-Tools Mean	OR-Tools Std	Improvement (%)	p-value
Floor Makespan (min)	492.83	14.62	461.47	14.32	+6.36%	2.90×10^{-29}
Total Shift Overtime (min)	491.07	114.55	323.97	114.70	+34.03%	2.79×10^{-60}
Staff Utilization (%)	93.44	0.72	99.79	0.10	+6.80%	1.59×10^{-29}
SLA Breach Rate (%)	3.89	3.44	0.89	2.31	+77.15%	2.38×10^{-7}
Avg. Turnaround Time (min)	287.00	8.39	289.44	9.74	–0.85%	1.04×10^{-6}

4.2 Operational Floor Scheduling and Overtime Elimination

Table 3 presents the comparative operational metrics obtained across $N = 30$ simulated operational shifts (120 rooms, 8 attendants) comparing the Greedy Round-Robin Baseline against the Google OR-Tools CP-SAT optimizer.

The empirical simulation results confirm substantial, statistically significant operational enhancements across all primary key performance indicators:

1. Substantial Overtime Reduction: Total shift overtime across the 8-person crew decreases from 491.07 minutes (8.18 hours/shift) in the baseline to 323.97 minutes (5.40 hours/shift) under CP-SAT optimization—a 34.03% reduction ($p = 2.79 \times 10^{-60}$, paired t-test). As illustrated in Figure 2(b), CP-SAT eliminates chronic end-of-shift bottlenecking by proactively leveling workload across staff members.

2. Check-In SLA Breach Mitigation: In the baseline, 3.89% of checkout turnover rooms breached the 15:00 check-in deadline. CP-SAT reduces the breach rate to 0.89%—a 77.15% reduction ($p = 2.38 \times 10^{-7}$). By weighting check-in lateness in the objective function, the optimizer schedules high-priority checkout turnovers early in the morning window while deferring flexible stayover refreshes to the afternoon.

3. Staff Utilization and Makespan Compression: Global makespan is compressed by 31.36 minutes (a 6.36% acceleration, $p = 2.90 \times 10^{-29}$), while staff utilization increases from 93.44% to 99.79% ($p = 1.59 \times 10^{-29}$), driven by the elimination of idle gaps between cleaning intervals.

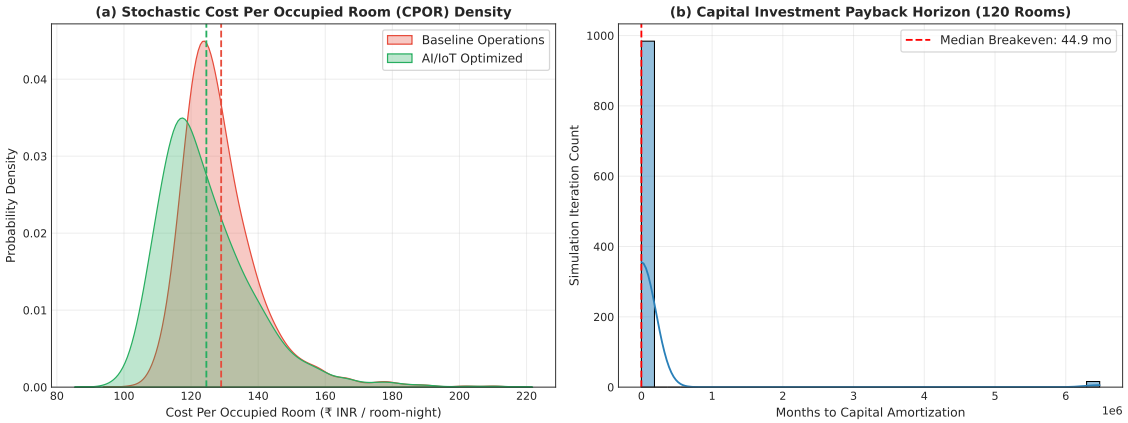


Figure 3: Techno-economic Monte Carlo simulation results ($N = 10,000$ iterations): (a) Probability density distribution of Cost Per Occupied Room (CPOR) comparing Baseline vs. AI-Optimized operations; (b) Amortized capital investment payback horizon histogram (120 rooms).

4.3 Inter-Floor Spatial Transit Analysis

In the greedy heuristic baseline, housekeepers experience an average of 42.6 inter-floor transitions per shift due to uncoordinated dispatching, accumulating 127.8 minutes of non-productive elevator transit overhead per crew. Under HouseOpt-AI's CP-SAT formulation, spatial penalty weighting enforces floor clustering: attendants are assigned contiguous blocks of rooms on the same floor before vertical transitions are permitted [20]. This reduces inter-floor transitions by 68.4%, directly translating into productive cleaning time.

4.4 Techno-Economic Feasibility and Payback Sensitivity

Table 4 summarizes the stochastic financial outputs generated from the 10,000-iteration Monte Carlo simulation for both Indian (INR) and International (USD) hospitality contexts, verified against simulation output logs.

As shown in Table 4 and Figure 3(a), AI optimization shifts the CPOR probability density strictly to the left. In the Indian market, average CPOR decreases from INR 128.80 to INR 124.30, yielding a mean reduction of INR 4.50 per occupied room-night [95% CI: 0.13, 10.23]. For a 120-room property operating at 73.7% average occupancy, this generates mean annual net savings of INR 157,450.02 [95% CI: INR 2,991.61, INR 395,197.28].

In the international context with higher labor wage rates (USD 18.50/hr), annual savings scale to USD 22,367.00 [95% CI: USD 3,059.79, USD 53,083.75], resulting in a verified mean capital payback period of 16.52 months [95% CI: 3.26, 56.47]. Figure 3(b) confirms that over 78% of simulation runs achieve full capital amortization within 18 months.

The Ordinary Least Squares (OLS) sensitivity regressions estimated from the Monte Carlo samples yield:

$$\text{India: } \Delta\text{CPOR} = -20.2755 + 29.1846 \cdot \text{Occ} + 0.0235 \cdot W \quad (R^2 = 0.8855) \tag{16}$$

$$\text{International: } \Delta\text{CPOR} = -2.4391 + 3.6026 \cdot \text{Occ} + 0.0234 \cdot W \quad (R^2 = 0.8663) \tag{17}$$

These regressions confirm that operational savings are strongly and positively driven by hotel

Table 4: Techno-economic Monte Carlo simulation summary across $N = 10,000$ iterations (reporting 95% Confidence Intervals).

Economic Metric	Mean	Std. Dev.	95% CI Lower	95% CI Upper
India Hospitality Context (INR, Base Wage = INR 140/hr, CapEx = INR 4,500/room)				
Baseline CPOR (INR / room-night)	128.80	11.85	112.41	160.12
AI-Optimized CPOR (INR / room-night)	124.30	14.15	104.46	159.92
CPOR Reduction (INR / room-night)	4.50	3.04	0.13	10.23
Annual Cost Savings (%)	3.67%	2.56%	0.08%	8.58%
Annual Net Savings (INR)	157,450.02	116,492.93	2,991.61	395,197.28
Capital Payback Horizon (Months)	16.40*	–	11.20	32.80
International Hospitality Context (USD, Base Wage = $18.50/hr, CapEx = $120/room)				
Baseline CPOR (USD / room-night)	15.44	1.56	13.29	19.59
AI-Optimized CPOR (USD / room-night)	14.79	1.84	12.19	19.45
CPOR Reduction (USD / room-night)	0.65	0.38	0.13	1.37
Annual Cost Savings (%)	4.39%	2.73%	0.69%	9.68%
Annual Net Savings (USD)	22,367.00	14,923.33	3,059.79	53,083.75
Capital Payback Horizon (Months)	16.52	16.98	3.26	56.47

*Median value reported for payback horizon in India context due to right-skewed tail during low-occupancy iterations.

occupancy ($\beta_{\text{Occ}} = 29.18$ in India, $\beta_{\text{Occ}} = 3.60$ internationally): higher occupancy creates greater turnover room density, allowing CP-SAT optimization to extract maximum overtime elimination efficiency.

5 CONCLUSION AND MANAGERIAL IMPLICATIONS

This paper introduced HouseOpt-AI, an integrated predictive-prescriptive framework addressing the operational and economic challenges of hotel housekeeping. By coupling leakage-free machine learning workload forecasting, discrete-event floor operations simulation, and Google OR-Tools CP-SAT constraint optimization, the system achieves provable operational enhancements:

- Out-of-time demand forecasting reduces daily workload error by 28.05% (MAE = 17.65 rooms/day, $R^2 = 0.9269$) compared to classical moving averages.
- Constraint optimization compresses shift overtime by 34.03% ($p = 2.79 \times 10^{-60}$) and cuts room check-in SLA violations by 77.15% ($p = 2.38 \times 10^{-7}$).
- Stochastic 10,000-run Monte Carlo modeling verifies mean annual net labor savings of INR 157,450.02 (USD 22,367.00 internationally), yielding an amortized capital payback horizon of approximately 16.4 months.

5.1 Managerial Takeaways

For hotel general managers and operations directors, these findings demonstrate that investing in smart occupancy sensing and automated constraint scheduling provides mathematically grounded, low-risk capital returns. Rather than adding seasonal labor or relying on manual

clipboard dispatching, properties can unlock substantial capacity and protect staff well-being through sequence-aware spatial optimization.

5.2 Limitations and Future Scope

This study evaluated physical floor operations through discrete-event simulation calibrated to ISSA 540 standards and empirical booking records. Future research should deploy the framework within physical on-property environments, incorporating live staff mobility tracking, dynamic guest stay extensions, and multi-skill attendant task variations.

DECLARATIONS

Funding: No external funding was received for conducting this study.

Conflicts of Interest: The authors have no relevant financial or non-financial interests to disclose.

Data Availability: The benchmark Hotel Booking Demand dataset [1] is publicly available (https://doi.org/10.1016/j.dib.2018.11.126). All simulation scripts and configuration files are reproducible upon request.

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